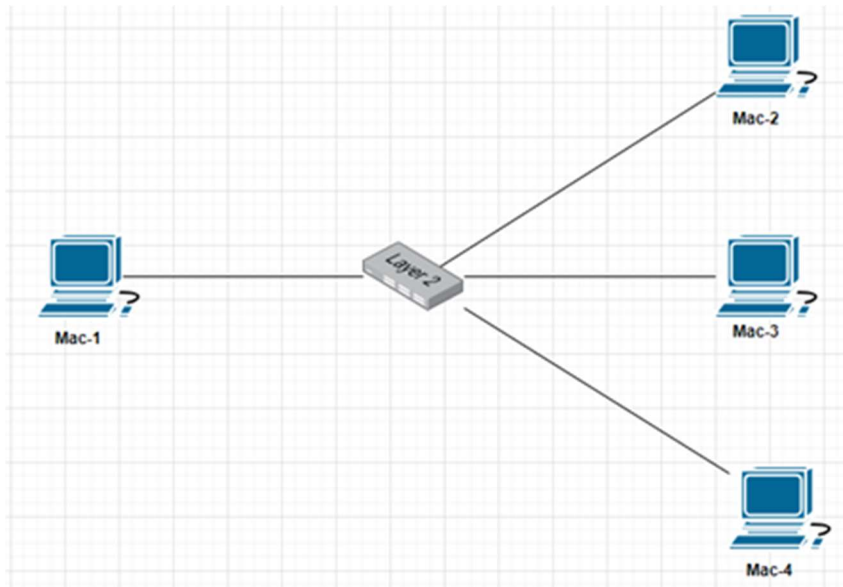


Topic: SSH Authentication with Public Keys

Aim:

1. Permit users on Mac-1 to log into Mac-2 using public keys
2. Restrict SSH authentication to public keys only
3. Transfer files with Rsync



Part 1A: Public Key authentication setup – Manual Method

On Mac-1

1. Login to Mac-1 as user: **netadmin**, Password: **ims**
2. Install SSH client: **apt install openssh-client**
3. On Mac-1 create SSH private and public key pairs
 - a. Open the Terminal
 - b. Confirm the logged in user is **netadmin** use the command: **whoami**
 - c. Run the following command and note down the IP address of Mac-1 (as Mac-1-IPAddress): **ip a**
 - d. Run the command (accept all prompts): **ssh-keygen**

On Mac-2

4. Read IP address of Mac-2
 - a. Login as user: **netadmin**, Password: **ims**
 - b. Open the terminal
 - c. Run the following command and note down the IP address of Mac-2 (as Mac-2-IPAddress): **ip a**
5. Install SSH server: **apt install openssh-server**
6. On Mac-2 create the ssh folder

- a. Run the following command (and accept prompts): **ssh-keygen**
- 7. On Mac-2 create the ssh authorized keys file
 - a. While still logged in as **netadmin**, run the command:
touch /home/netadmin/.ssh/authorized_keys

On Mac-1

- 8. Copy SSH public key from Mac-1 to Mac-2
 - a. On Mac-1, while still logged in as **netadmin**
 - b. Open a new terminal (You should have two terminal windows/tabs open)
 - c. SSH to **Mac-2** use the following command (You will be asked for a password for netadmin on Mac-2): **ssh netadmin@Mac-2-IPAddress**
 - d. **Note:** The first terminal window (Window 1) is logged in to Mac-1 and second terminal (window 2) is logged into Mac-2 (You can confirm this by checking the IP address¹)
 - e. In window 2 (Mac-2), open the authorized keys file: **nano /home/netadmin/.ssh/authorized_keys²**
 - f. In window 1 (Mac-1), open the public key file: **cat /home/netadmin/.ssh/id_rsa.pub³**
 - g. Copy and paste the string of the public key in (f) into the authorized_keys file in (e)
 - h. Save the file on Mac-2 with **Ctrl+O**
 - i. Close the open files on Mac-1 and Mac-2 with **Ctrl+x**
 - j. Close window 2 using **Ctrl+d**
- 9. Testing authentication by public keys from Mac-1 to Mac-2
 - a. On Mac-1, close all terminals and open a new terminal
 - b. SSH from Mac-1 to Mac-2 as netadmin: **ssh netadmin@Mac-2-IPAddress**
- 10. Answer the following questions
 - a. Did you get prompted for the password?
 - b. If yes in (a) check if you missed a step
 - c. If no in (a), explain what happened.

Part 1B: Public Key authentication setup – Automatic Method

On Mac-3

- 11. Install SSH server: **apt install openssh-server**
- 12. Read IP address of Mac-3
 - a. Login as user: **netadmin**, Password: **ims**
 - b. Open the terminal
 - c. Run the following command and note down the IP address of Mac-2 (as Mac-3-IPAddress): **ip a**

On Mac-1

- 13. Export the Mac-1's SSH public key to Mac-3

¹ This procedure is very important when working with remote computers

² File names might differ based on the Linux distribution

³ File names might differ based on the Linux distribution

- a. Open a terminal on Mac-1 (Ensure you are logged in as netadmin)
 - b. Run the following command to export the public key to Mac-3 (You will be prompted for a password): **ssh-copy-id Mac-3-IP-Address**
- 14. Testing authentication by public keys from Mac-1 to Mac-3
 - a. On Mac-1, close all terminals and open a new terminal
 - b. SSH from Mac-1 to Mac-3 as netadmin: **ssh netadmin@Mac-3-IPAddress**
- 15. Answer the following questions
 - a. Did you get prompted for the password?
 - b. If yes in (a) check if you missed a step
 - c. If no in (a), explain what happened.
- 16. Export Mac-1 public key to Mac-4
 - d. Repeat steps 11 to 13. Replace Mac-3 with Mac-4 appropriately.

Part 2: File Transfer using Rsync

On Mac-1

- 17. Install rsync: **apt install rsync**

On Mac-2

- 18. Install rsync: **apt install rsync**

On Mac-1

- 19. Download some files from Internet
 - a. On Mac-1 download 8 picture files (e.g. cars, celebs, animals etc) and 2 document files (i.e. one .ppt and one .doc)
 - b. Save them in a folder named MyFiles (i.e. in /home/netadmin/MyFiles/)
- 20. Copy the downloaded files to Mac-2
 - a. On Mac-1 run the command: **rsync -avz /home/netadmin/MyFiles netadmin@Mac-2-IPAddress:**
 - b. Capture a screenshot for submission
 - c. Explain your observation

On Mac-2

- 21. Confirm files have been transferred
 - a. On Mac-2, run the command: **ls -l /home/netadmin/MyFiles**
 - b. Were all files copied over from Mac-1?

On Mac-1

- 22. Differential backup
 - a. Open and edit the .doc file (and save and close it)
 - b. Repeat step (20), capture a screenshot and explain your observation
 - c. Were all files copied over again to Mac-2?